

SUNCLAVE · DESIGN DECISION RECORD

SunClave TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 5 and 12 decided by Amish on 2026-10-02 (recommendations approved, SCL-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 4 and 6 to 11; items 13 to 18 accepted later the same day, see SCL-DDR-002); items 5 and 12 were decided on 2026-10-02 (SCL-DEC-001)

Safety: SunClave is a research and educational prototype, not a medical device. It concentrates sunlight to a burning intensity and holds steam at about 121 °C in a pressure vessel. Item 5 (drilling the cooker lid) is a pressure-safety trade-off and is deliberately left open.

■ Context

The TRL 2 review note (`docs/REVIEW.md` , session 2026-09-25, /populate) listed twelve items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction approved cross-cutting SwapCell interface items, a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 section) and SCL-PRC-001 v0.2. They are not repeated here, except for the open items.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Altitude strategy	Decided by Amish, 2026-09-25: go with recommendation. Option B: keep the 103.4 kPa cooker and extend the hold by an equivalent-exposure calculation ($z = 10\text{ }^{\circ}\text{C}$), recorded plainly as a lower-temperature cycle and never labeled 121 $^{\circ}\text{C}$, with a parallel search for a maker-rated vessel of about 130 kPa gauge (option A)	SCL-REQ-001 v0.3 R1 (redefined), SCL-CAL-001 section 2
2	Budget	Decided by Amish, 2026-09-25: go with recommendation. Option (b), then (a) if the savings fall short. The cuts (timber stand, OLED instead of e-paper, a USB power bank instead of the logger panel and cell) save about \$29, short of the \$37 needed, so <code>budget_usd</code> is raised from \$400 to \$450	<code>project.yaml</code> , <code>bom/bom.csv</code> , SCL-REQ-001 R13 and R17 (redefined)
3	Pitch wording	Decided by Amish, 2026-09-25: go with recommendation. The pitch now reads "with a calibrated temperature, pressure and time logger that records every cycle" instead of "a validated temperature and time logger to prove each cycle"	<code>project.yaml</code> , <code>README.md</code>
4	Pressure vessel	Decided by Amish, 2026-09-25: go with recommendation. A household 12 L aluminum cooker, with a purpose-made sterilizer such as the All-American 1915X only as a reference vessel if a partner already owns one	SCL-PRC-001 v0.3, <code>bom/bom.csv</code> item 6
6	Level cooker at the focus	Decided by Amish, 2026-09-25: go with recommendation. The cooker stays level at the focus and the dish tilts about it. SCL-CAL-001 section 10 shows the holder must be fixed to the stand (a holder swinging on the axis would be top-heavy); this implements the decision and does not change it	<code>cad/src/model.py</code> item 5
7	Concentrator type	Decided by Amish, 2026-09-25: go with recommendation. Build the 1.4 m petal dish; use a bought SK14 if the partner already has one	SCL-PRC-001 v0.3
8	Aiming	Decided by Amish, 2026-09-25: go with recommendation. Manual aiming with a shadow gnomon; no tracker	SCL-PRC-001 v0.3
9	Load limits	Decided by Amish, 2026-09-25: go with recommendation. Solid instruments only, unwrapped or single-wrapped, as a firm limit	SCL-REQ-001 R3
10	Logger pass criteria	Decided by Amish, 2026-09-25: go with recommendation. Purge plateau of 5 min or more, hold temperature and time met (the equivalent hold under item 1), and measured temperature within 2 $^{\circ}\text{C}$ of saturation throughout the hold	SCL-REQ-001 R1, R6 and R12
11	Backup heat	Decided by Amish, 2026-09-25: go with recommendation. The same cooker on a wood, charcoal or LPG stove on cloudy days, logged the same way	SCL-PRC-001 v0.3 concept of operation

The problem line had no recommended rewording in the TRL 2 review and is unchanged.

Cross-cutting approvals from the same instruction, recorded here:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell interface adds a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles. SunClave does not use SwapCell (the logger runs from a USB power bank), so no v0.3 items apply.

- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting). Not applicable: SunClave has no SwapCell pack.
- **Co-design partners per area later.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. Item 12 stays open under this rule.

Items that remain open

Table 2. Items left open by this record. Items 13 to 18 were later decided by Amish on 2026-09-25 (SCL-DDR-002); items 5 and 12 were decided on 2026-10-02 (SCL-DEC-001).

#	Item	Status and options
5	Mounting the gauge, relief valve and probe gland: (i) drill the maker's lid with reinforcing washers, or (ii) a cooker with factory ports, or an adapter plate on the regulator stem	Decided by Amish, 2026-10-02 (recommendation approved, SCL-DEC-001): the maker's lid is not drilled; a pressure canner sold with a factory gauge and relief valve is used, and the probe gland goes on an adapter plate on the vent stem that keeps a bore of 3 mm or more and leaves the overpressure plug untouched; a lid is drilled only with the maker's written approval.
12	First partner and site for co-design	Decided by Amish, 2026-10-02 (recommendation approved, SCL-DEC-001): first partner to approach is a university biomedical or global health engineering group with an established rural clinic partner in a sunny region below 2,400 m, for example through an Engineering World Health university chapter.
13	R14 tracking: power stays within 10 % of on-target for about 13 min of sun motion, not 15 min	Options: (a) relax R14 to retargeting every 12 min, with a logger reminder; (b) widen the black band on the wall to catch more of the drifting spot, at the cost of more heat loss; (c) hold the petals to a tighter slope error. Recommendation: (a). Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)
14	R16 mass: about 41.0 kg against 40 kg, because of the decided timber stand	Options: (a) relax the total to 45 kg and keep the 20 kg piece limit (met at 19.4 kg); (b) lighter timber sections; (c) return to a steel stand (about \$15 more). Recommendation: (a). Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)
15	R11 guard and hot surfaces: no focal-zone guard is defined, and the black base and band (above 60 °C) are within reach	Options: (a) restate R11 so the whole pressure vessel and its fittings count as a hot zone like the lid, and rely on turning the dish off the sun before reaching in (R9), the parking cover, goggles and a marked keep-out on the ground; (b) design a physical guard at a later revision. A guard cannot close the converging light cone without shading the dish. Recommendation: (a). Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)
16	R15 stability: tipping factor 1.30 at 15° elevation, and the 153 N wind force exceeds the grip of two locked castors	Options: (a) four locking castors (about \$4 more, within the \$20 margin); (b) wheel chocks; (c) ballast. Recommendation: (a), plus a rule to park the dish face-up in high wind. Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)
17	R10 at altitude: option B holds in strong sun can leave less than 0.5 L (0.14 kg at 1,800 m and 1,000 W/m ²)	Options: (a) an operating rule to trim the dish off the sun during holds so only a little steam vents, with a water-budget warning in the logger; (b) 2.0 L of water above 1,000 m, which leaves only about 8 mm between the water and the basket. Recommendation: (a). Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)
18	R12 air-removal check at risk (0.89 K uncertainty against the 2 K threshold)	Option: a 0 to 300 kPa absolute transducer (0.64 K) instead of 0 to 500 kPa, with an overpressure rating above the relief set point. Recommendation: adopt it. Decided by Amish, 2026-09-25: go with recommendation (SCL-DDR-002)

■ Consequences

- SCL-PRB-001, SCL-PRC-001 and SCL-REQ-001 move to v0.3 with these decisions; the precis no longer lists items 1 to 4 and 6 to 11 as proposed.
- SCL-REQ-001 R1 is restated as an equivalent exposure with the real temperature recorded, R13 as a USB-recharged power bank, and R17 as \$450.
- The TRL 3 model, drawing SCL-DWG-001, `bom/bom.csv` and SCL-CAL-001 follow the decided vessel, holder, dish, aiming, timber stand, OLED logger and power bank.
- Items 13 to 18 were decided later on 2026-09-25 and are implemented through SCL-DDR-002. Items 5 and 12 do not change the model or the BOM until Amish decides them.
- TRL 4 work (building, testing, purchasing) is on hold by Amish's instruction, whatever the outcome of the open items.